

ASSIGNMENT: RADAR

CASE STUDY:
VOLCANIC ERRUPTION IN ULAWUN, PAPUA NEW GUINEA

SUBMITTED BY:

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1. Study Area Selection

Ulawun is selected as the study area which is located in West New Britain Province, on the island of New Britain, Papua New Guinea. Ulawun is one of the most active volcanoes in Papua New Guinea which have caused widespread ashfall, damage and evacuations of thousands of residents.

The volcano eruption was recorded at approximately 7am on 26 June 2019, that was quickly upgraded to a major sub-Plinian eruption, with ash climbing to 19 kilometers approximately.

Ulawun

 23 languages 

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From Wikipedia, the free encyclopedia

Coordinates:  5°03′00″S 151°20′00″E﻿ / ﻿



This article includes a [list of references](#), [related reading](#), or [external links](#), **but its sources remain unclear because it lacks [inline citations](#)**. Please help [improve](#) this article by [introducing](#) more precise citations. *(August 2019)* ([Learn how and when to remove this message](#))

Ulawun is a [basaltic](#) and [andesitic stratovolcano](#) in [West New Britain Province](#), on the island of [New Britain](#) in [Papua New Guinea](#).

About 130 km (81 mi) southwest of the township of [Rabaul](#), Ulawun is the highest mountain in [New Britain](#) and the second in the [Bismarck Archipelago](#) at 2,334 metres (7,657 ft), and one of the most active volcanoes in Papua New Guinea. A total of 22 recorded eruptions have occurred since the 18th century; the first, in 1700, was recorded by [William Dampier](#). Several thousand people live near the volcano. Because of its eruptive history and proximity to populated areas, Ulawun is considered one of the [Decade Volcanoes](#).

Recent activity [\[edit \]](#)

The last few years have seen almost constant activity at Ulawun, with frequent small explosions, and have caused great damage and loss of life.

The volcano erupted at approximately 7am on 26 June 2019, and was quickly upgraded to a major [sub-Plinian](#) eruption, with ash climbing to 19 kilometres (62,000 ft).^[2] Over 5000 people were evacuated, and flights into nearby [Hoskins Airport](#) were cancelled. Lava cut the New Britain Highway in three different locations.^[3]

On 20 November 2023 ash emissions and eruption noises lead to the alert level being raised to stage 3. The eruption intensified the next hours and produced a plume that reached at an altitude of 15 kilometres (49,000 ft)^[4] that drifted 520 kilometres (320 mi) west, and alert level was raised to 4, the top of the scale. [Pyroclastic flows](#) descended the northwest and south flank and ashfall was recorded in the north and northwest of the volcano. The alert level was lowered to

Ulawun



Ulawun issuing passive steaming

Highest point

Elevation 2,334 m (7,657 ft)^[1]

Prominence 2,334 m (7,657 ft)

Listing Ultra

Coordinates  5°03′00″S 151°20′00″E﻿ / ﻿

Geography

Ulawun is located in  [Ulawun](#)
[Papua New Guinea](#)

Location [West New Britain Province](#), [Papua New Guinea](#)

Geology

Mountain type [Stratovolcano](#)

Volcanic arc [Bismarck volcanic arc](#)

Last eruption 2023

2. Data Download

Event was occurred in 2019-06-26. The pre-event (**2019-06-22**) and post-event (**2019-07-04**) product were downloaded from ASF Data Search Vertex platform which offers efficient and robust access for discovering and downloading SAR data. For this, area of interest was defined with polygon geometry and search the data by defining required parameters such as data filters, **L1 SLC** as file type since we need phase data, **IW** as beam mode, **VV** as polarization and **ascending** flight direction.

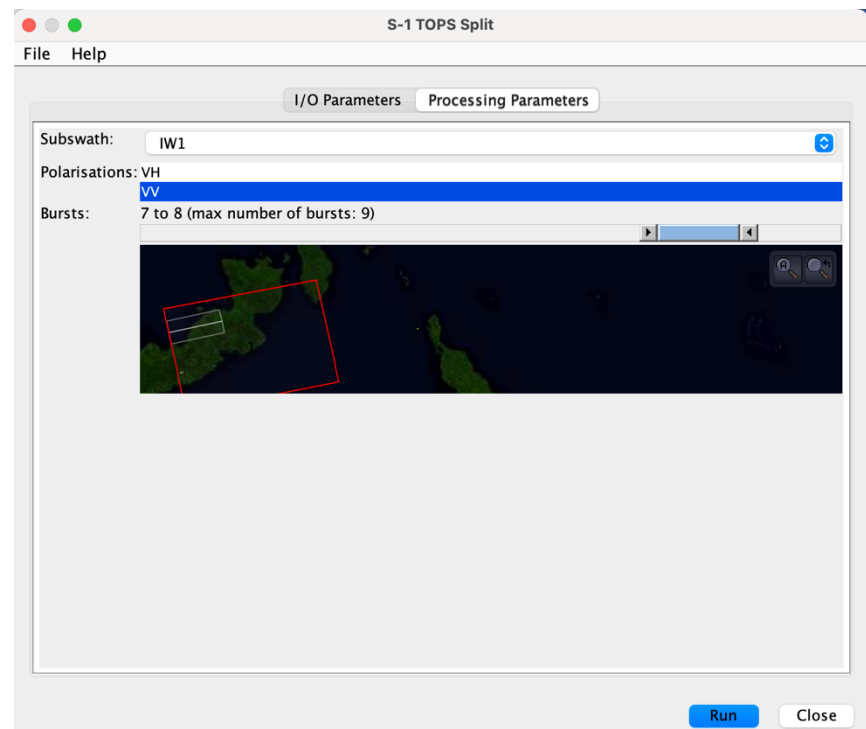
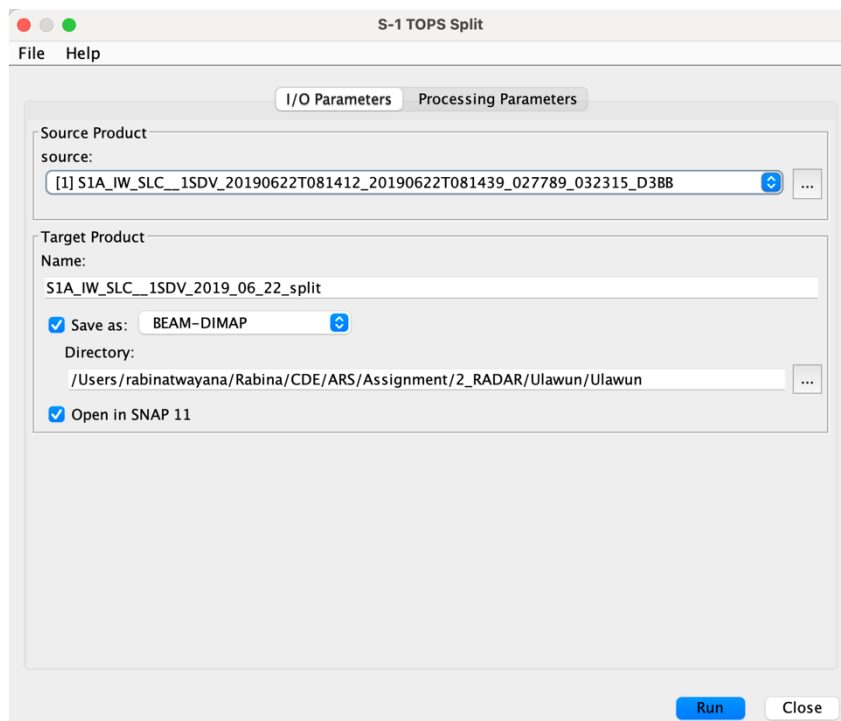
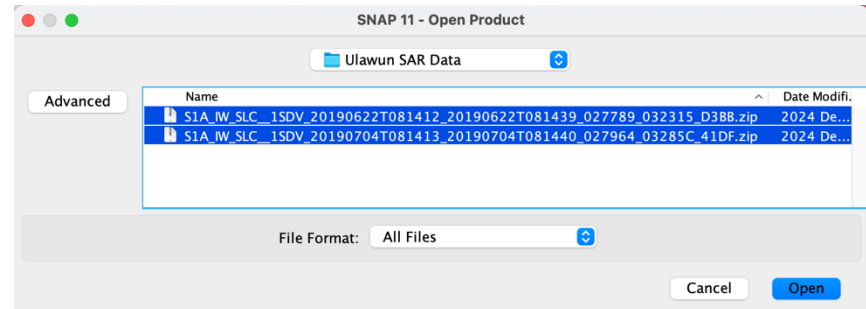
The screenshot displays the ASF Data Search Vertex web interface. The top navigation bar includes the NASA EarthData logo, a dropdown for 'Other DAACs', and a 'Feedback' link. The main search area features a 'Search Type' dropdown set to 'Geographic Search', a 'Dataset' dropdown set to 'Sentinel-1', and an 'Area of Interest - WKT' field containing a polygon geometry: `POLYGON((151.2483 -5.0`. A 'SEARCH' button is present, and a status bar indicates '250 of 5,018 Files'. The map view shows a satellite image of New Britain with a yellow polygon area of interest. The map controls include 'Map View', 'Zoom', 'Layers', 'Area Of Interest', and 'Opacity' (set to 100%). The search filters panel on the right includes 'Area of Interest Options' with a 'Drag and Drop' section for geospatial files, 'Date Filters' with 'Start Date' (6/20/2019) and 'End Date' (7/17/2019), and 'Additional Filters' with dropdowns for 'L1 Single Look Compl.' (1/14 file types selected), 'IW' (1/9 beam modes selected), 'VV' (1/8 polarizations selected), 'Ascending' (1/2 flight directions selected), 'Subtype' (0/2 subtypes selected), and 'Group ID'. The 'Downloads' section at the bottom lists two files: 'S1A_IW_SLC_1SDV_20190704T081413_20190704T081440_0_410F - 3.91 GB' and 'S1A_IW_SLC_1SDV_20190622T081412_20190622T081439_0_D3BB - 3.84 GB', both identified as 'L1 Single Look Complex (SLC)'. A 'Clear' button and links for 'Copy File IDs', 'Copy URLs', 'Data Download', and 'Metadata Download' are at the bottom.

3. Preprocessing

3.1. Split data to region of interest

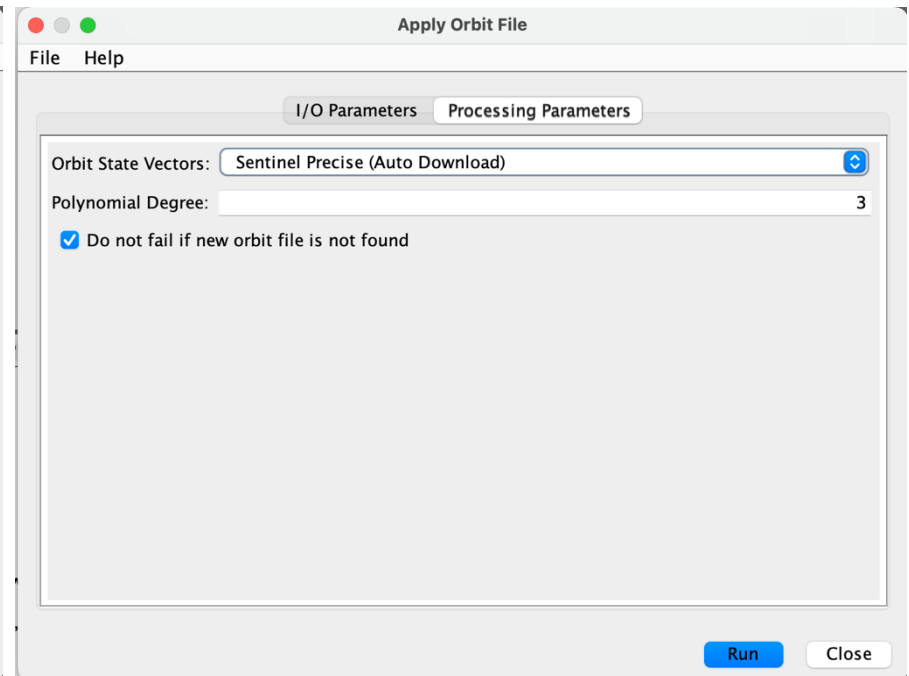
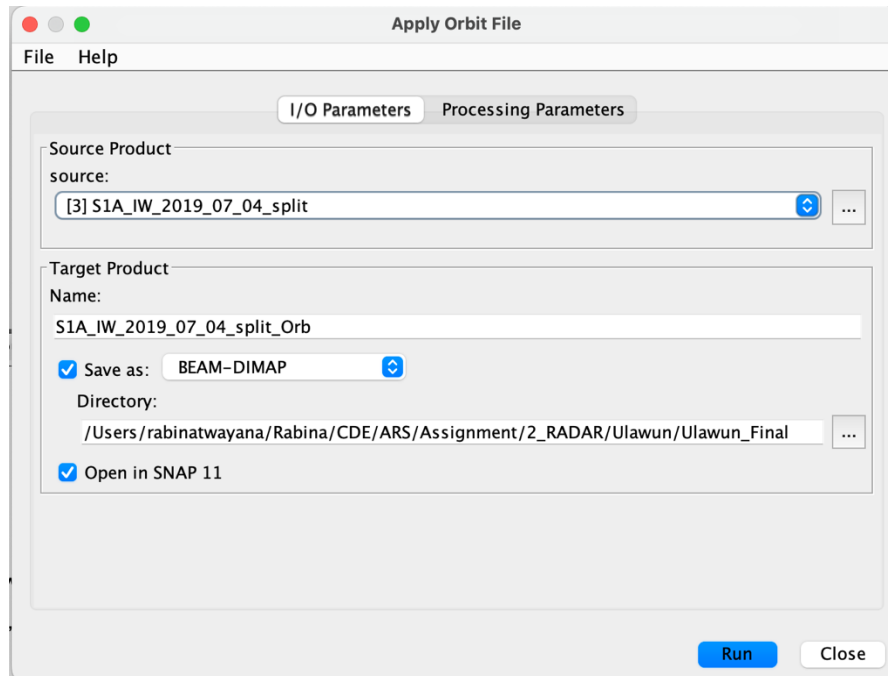
This is the first processing steps in InSAR processing chain. The downloaded products in .zip format were imported in SNAP tool (File >> Import product). The downloaded data consist of three sub-swath (IW1,IW2,IW3) but the study area falls under certain swath. To select the study area and reduce the size of image, this operation is carried out.

Radar > Sentinel-1 TOPS > S-1 Tops Split operator was used apply to splits in sub swath IW1 with bursts 7 and 8 that covered the study area and only VV is selected as polarisations parameter. This operation is carried out for both product.



3.2. Orbit Correction

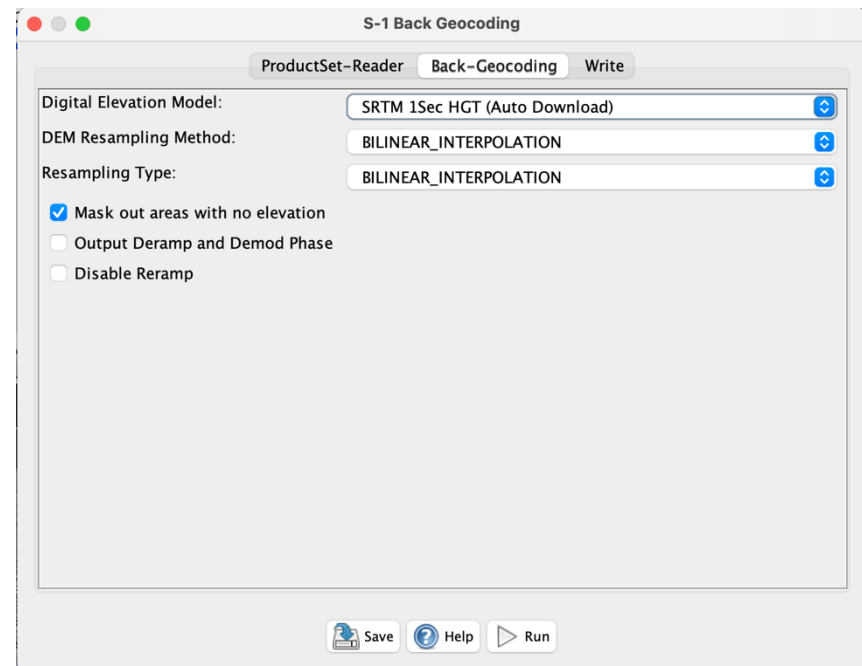
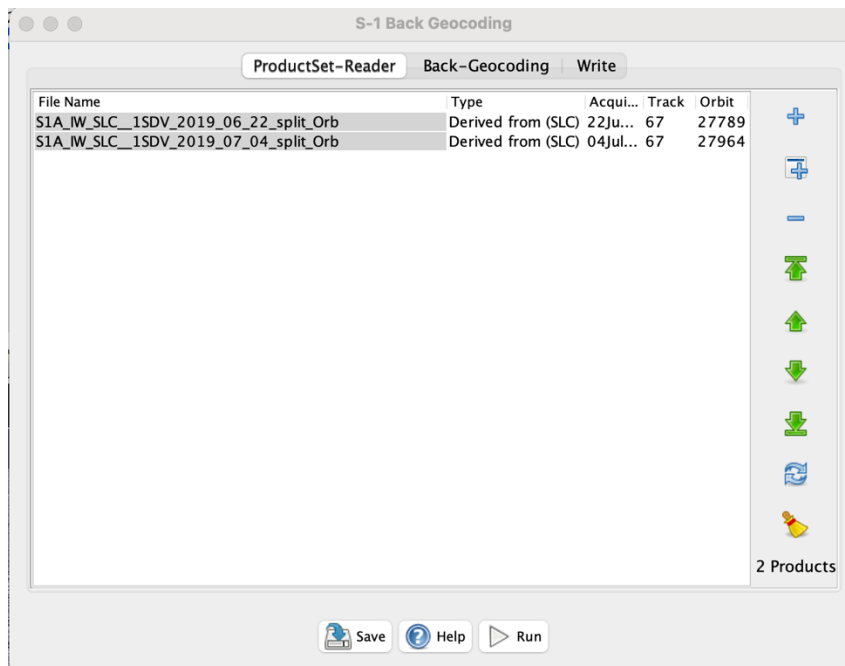
Orbital correction is applied to account for the orbital path based on Orbit State Vectors (OSV) file. This operation (**Radar > Apply Orbit File**) is also applied for the product obtain from S-1 TOPS step. If analysis is carried out for the recent event less than two weeks, "Do not fail if new orbit file is not found" option need to be checked.



4. Generation of Topographic Interferogram

4.1. Back Geocoding

The S-1 Back Geocoding operator (Radar > Coregistration > S-1 Tops Coregistration > S1- Back Geocoding) is carried out to co-registers the two split products based on the orbit information and digital elevation model (DEM). For this, both the product from orbit correction is selected and DEM (SRTM 1Sec HGT) was selected which is downloaded automatically by SNAP during processing. From this operation one stack product is obtained.



4.3. Enhanced Spectral Diversity

This operation (Radar > Coregistration > S1 TOPS Coregistration > S-1 Enhanced Spectral Diversity) is carried out to performs the joint coregistration of a stack image obtained from geocoding. All the processing parameters were used as default in this step.

The screenshot shows the 'I/O Parameters' tab of the 'S-1 Enhanced Spectral Diversity' dialog. It contains fields for 'Source Product' (set to '[7] S1A_IW_SLC__1SDV_2019_06_22_split_Orb_Stack1'), 'Target Product Name' (set to 'S1A_IW_SLC__1SDV_2019_06_22_split_Orb_Stack1_esd'), 'Save as' (set to 'BEAM-DIMAP'), 'Directory' (set to 'itwayana/Rabina/CDE/ARS/Assignment/2_RADAR/Ulawun/Ulawun'), and a checked 'Open in SNAP 11' option. 'Run' and 'Close' buttons are at the bottom.

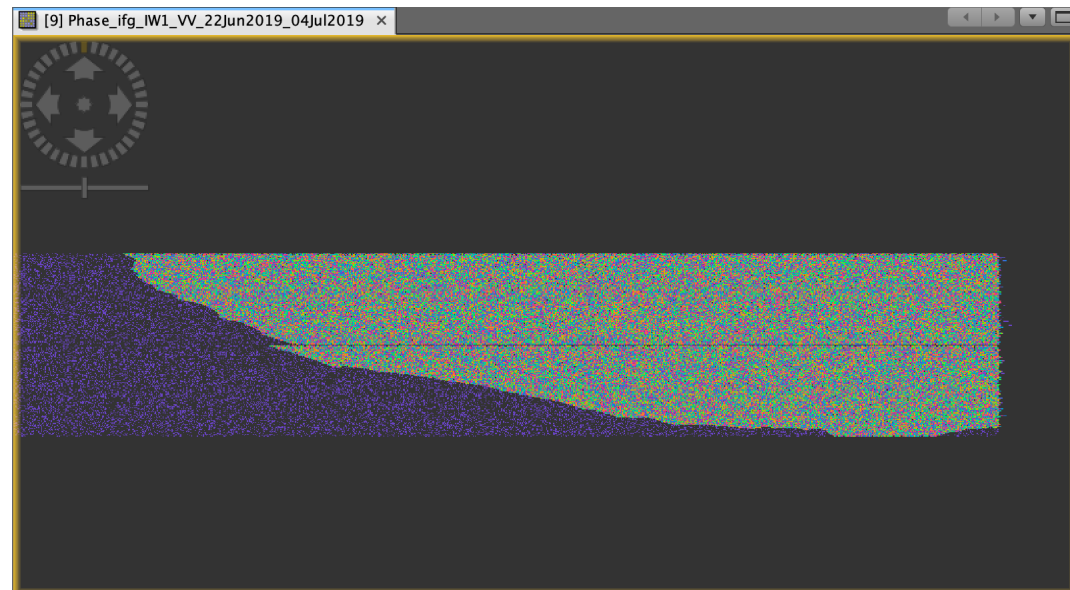
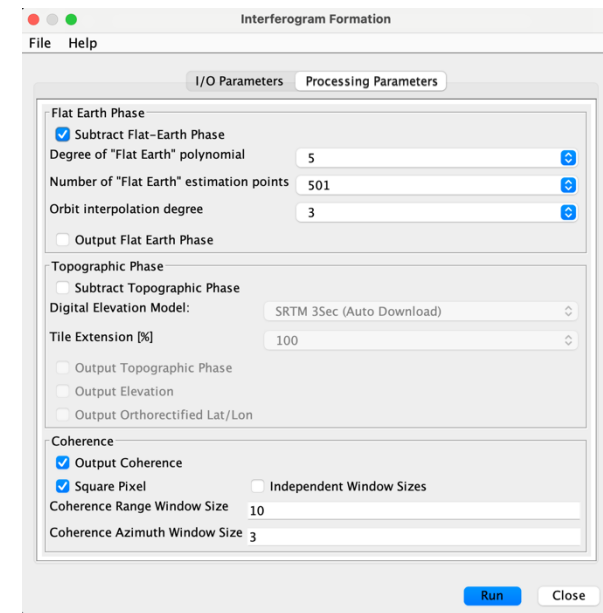
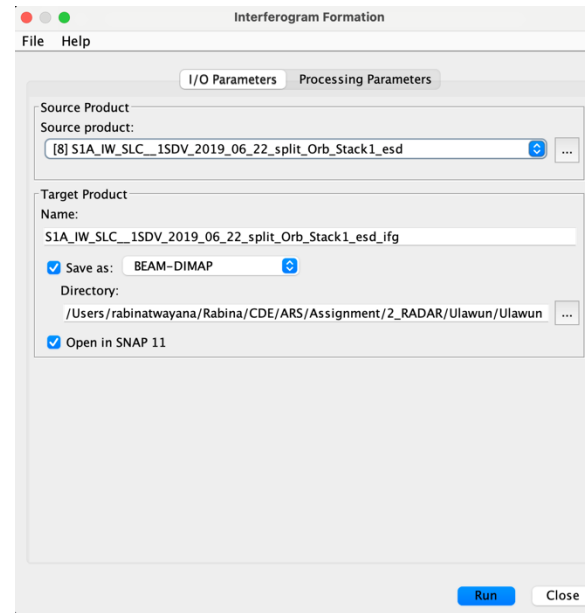
Parameter	Value
Source Product	[7] S1A_IW_SLC__1SDV_2019_06_22_split_Orb_Stack1
Target Product Name	S1A_IW_SLC__1SDV_2019_06_22_split_Orb_Stack1_esd
Save as	BEAM-DIMAP
Directory	itwayana/Rabina/CDE/ARS/Assignment/2_RADAR/Ulawun/Ulawun
Open in SNAP 11	☒

The screenshot shows the 'Processing Parameters' tab of the 'S-1 Enhanced Spectral Diversity' dialog. It contains various numerical and categorical parameters for registration and processing, all set to their default values. 'Run' and 'Close' buttons are at the bottom.

Parameter	Value
Registration Window Width	512
Registration Window Height	512
Search Window Accuracy in Azimuth Direction	16
Search Window Accuracy in Range Direction	16
Window oversampling factor	128
Cross-Correlation Threshold	0.1
Coherence Threshold for Outlier Removal	0.3
Number of Windows Per Overlap for ESD	10
ESD Estimator	Periodogram
Weight function	Inv Quadratic
Temporal baseline type	Number of images
Maximum temporal baseline (inclusive)	4
Integration method	L1 and L2
Do not write target bands (store range and azimuth offsets in json files)	☐
Use user supplied range shift (please enter it below)	☐
The overall range shift in pixels	0.0
Use user supplied azimuth shift (please enter it below)	☐
The overall azimuth shift in pixels	0.0

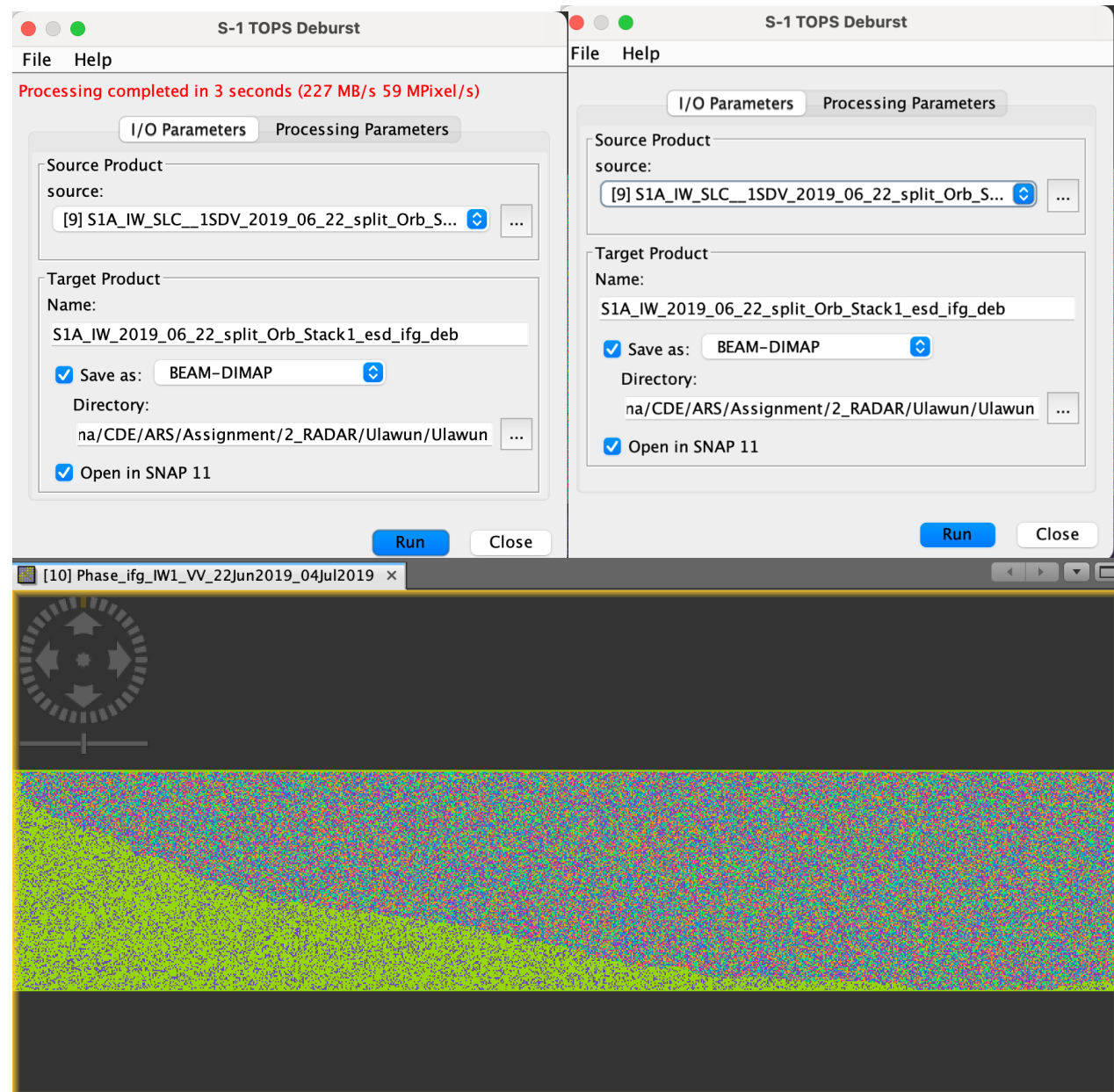
4.3. Interferogram Formation

This step (Radar > Interferometric > Products > Interferogram Formation) is carried out to calculate the phase difference between two SAR images. This difference is stored as an interferogram, which visually represents phase changes across the imaged area.



4.4. TOPSAR Deburst

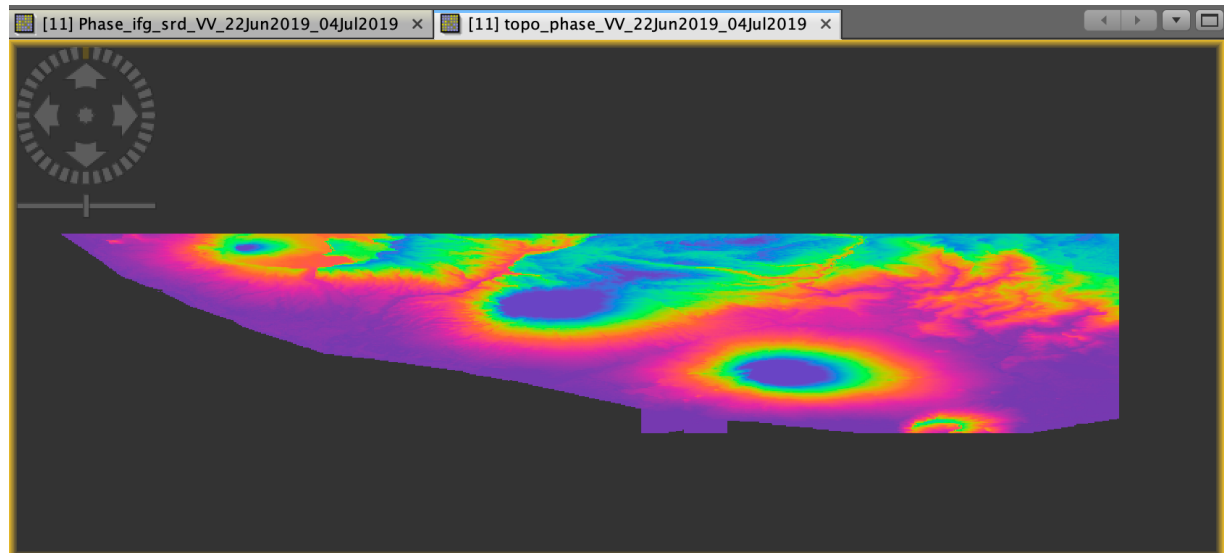
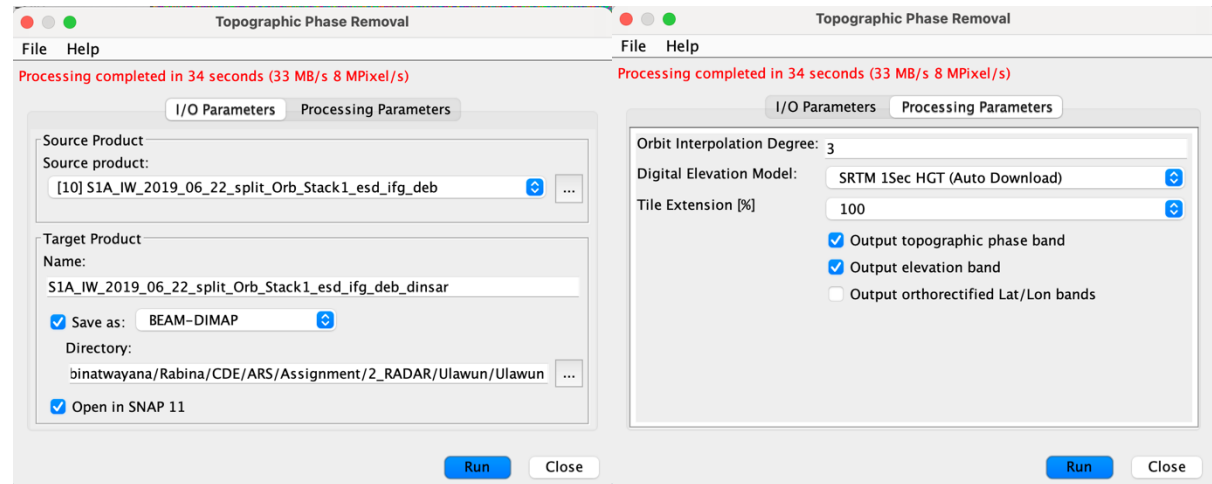
This operation (RADAR > Sentinel 1 - TOPS > S-1 TOPS-Deburst) is carried out to merge the individual bursts of SAR image in TOPS mode into a seamless, continuous image. All the default parameters were used in this process.



5. Generation of Differential Interferogram

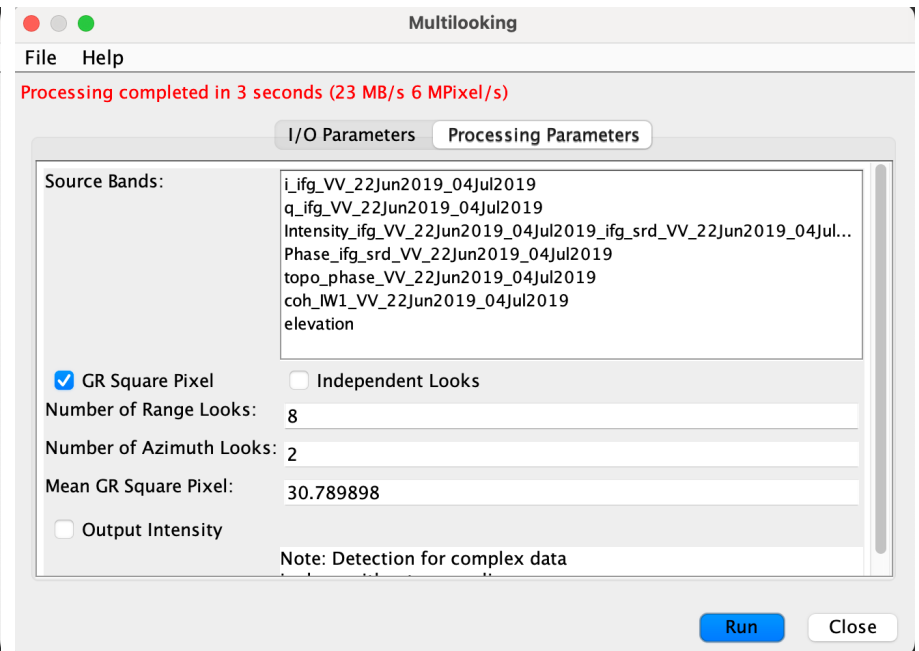
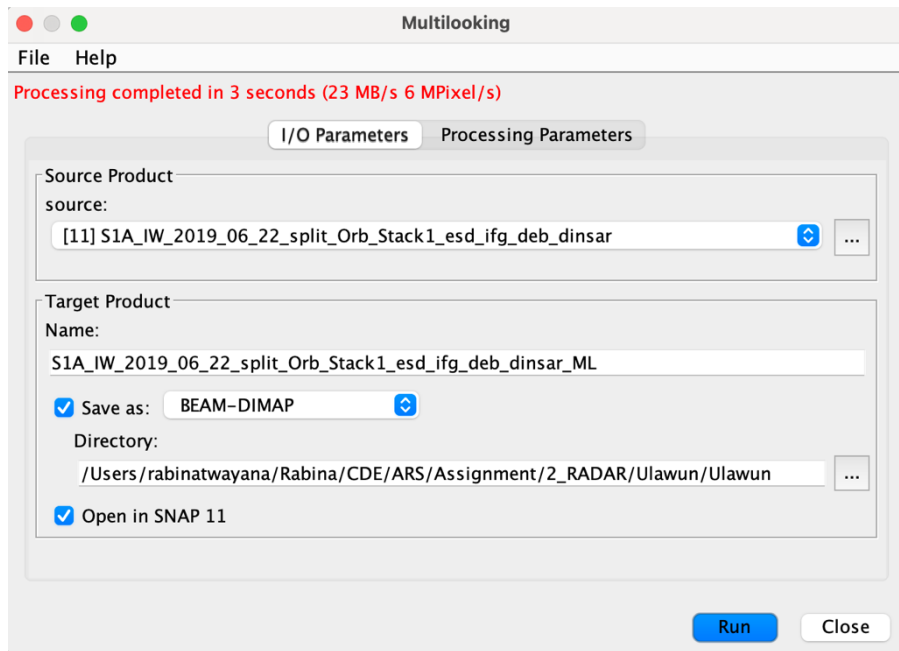
5.1. Topographic Phase Removal

This operation (Radar > Interferometric > Products > Topographic Phase Removal) is carried out to correct the phase shift caused topographic features based on DEM. The same DEM source is selected in this step as before and checked the options **Output topographic phase band** and **Output elevation band**.

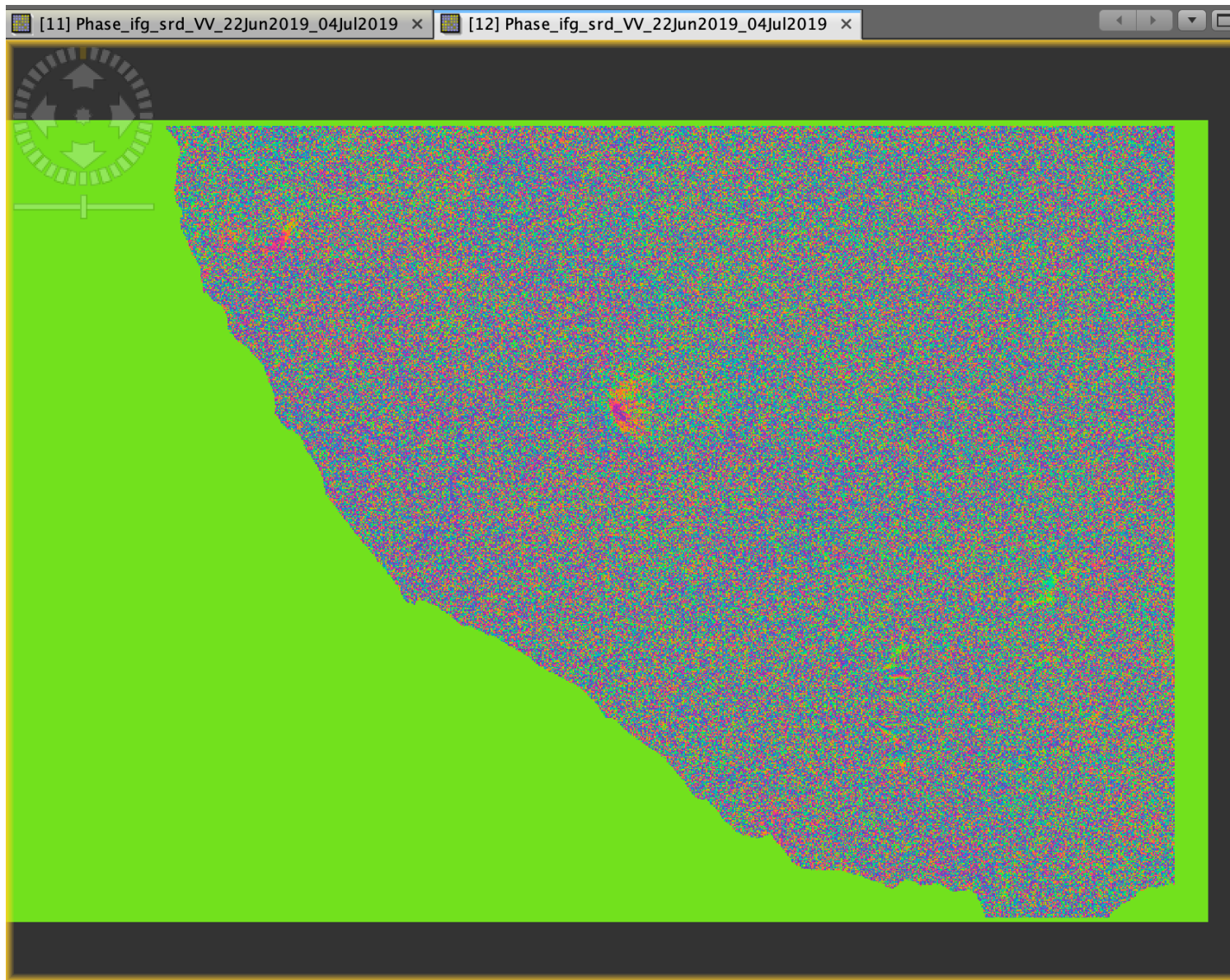


5.2. Multilooking

This option (RADAR > SAR utilities > Multilooking) is carried out to make the uniform square pixel and to set the desired resolution. In this case, number of range look is set as 8 obtain the pixel of approximately 30m.

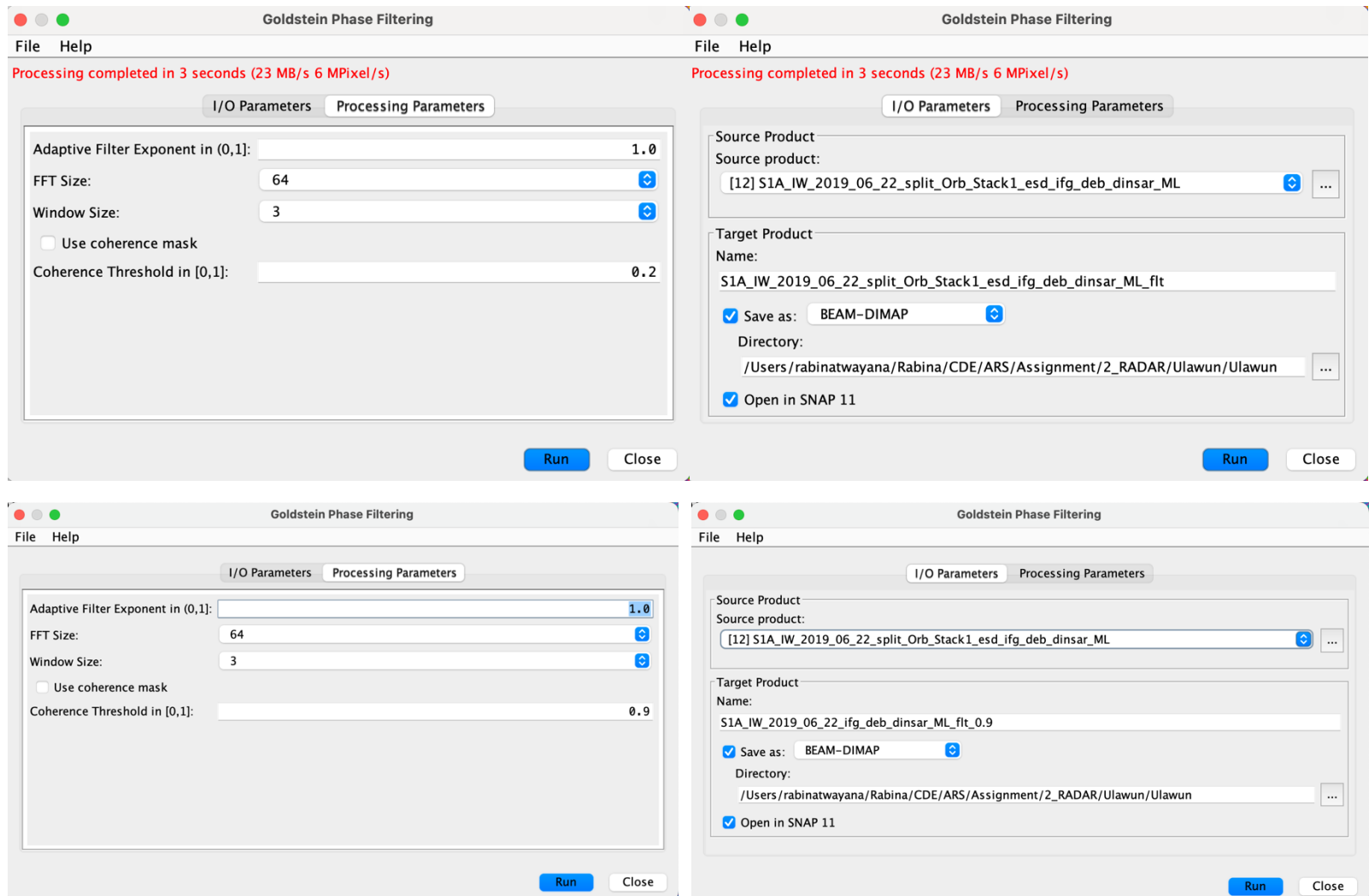


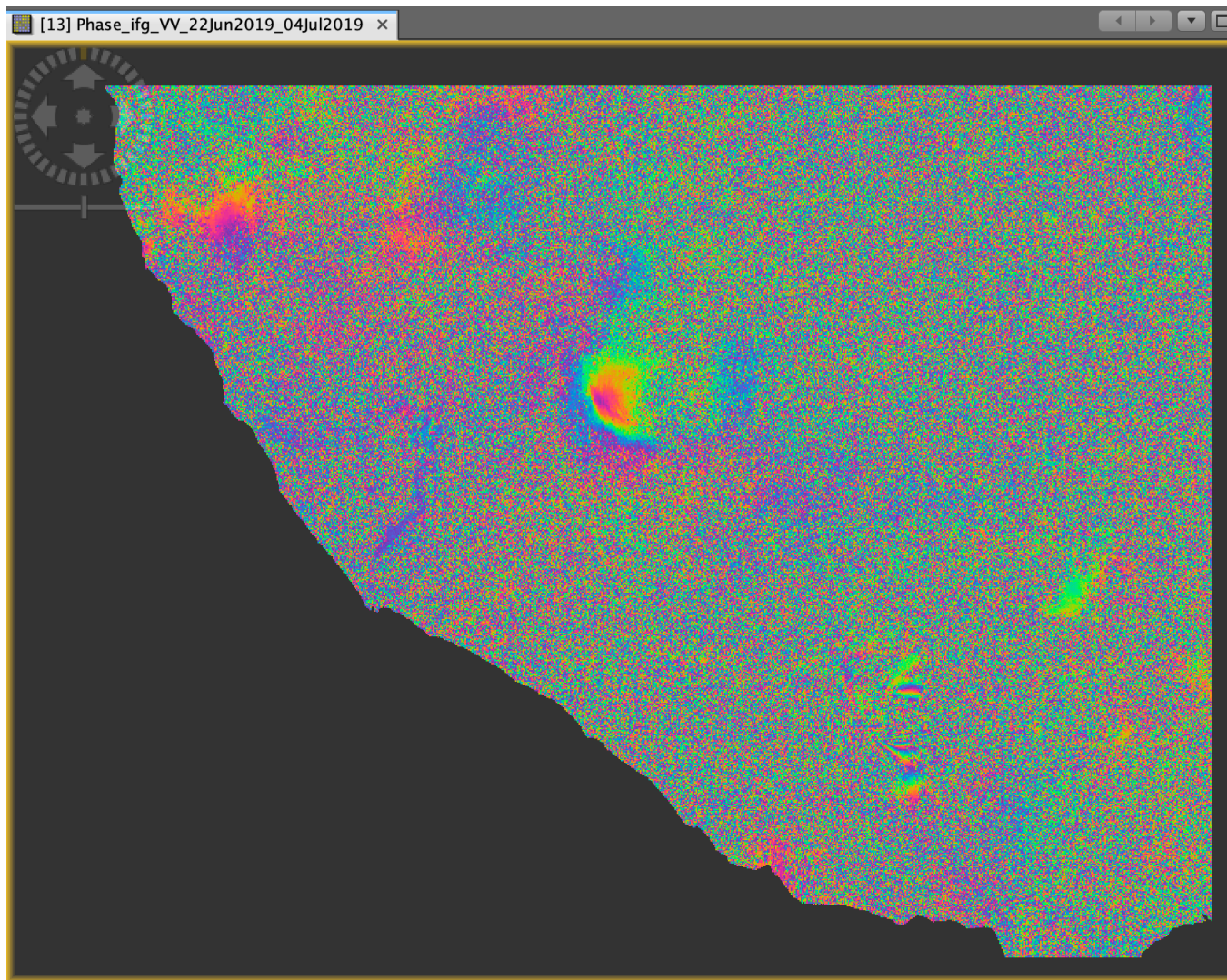
This is the output of after multilooking operation.



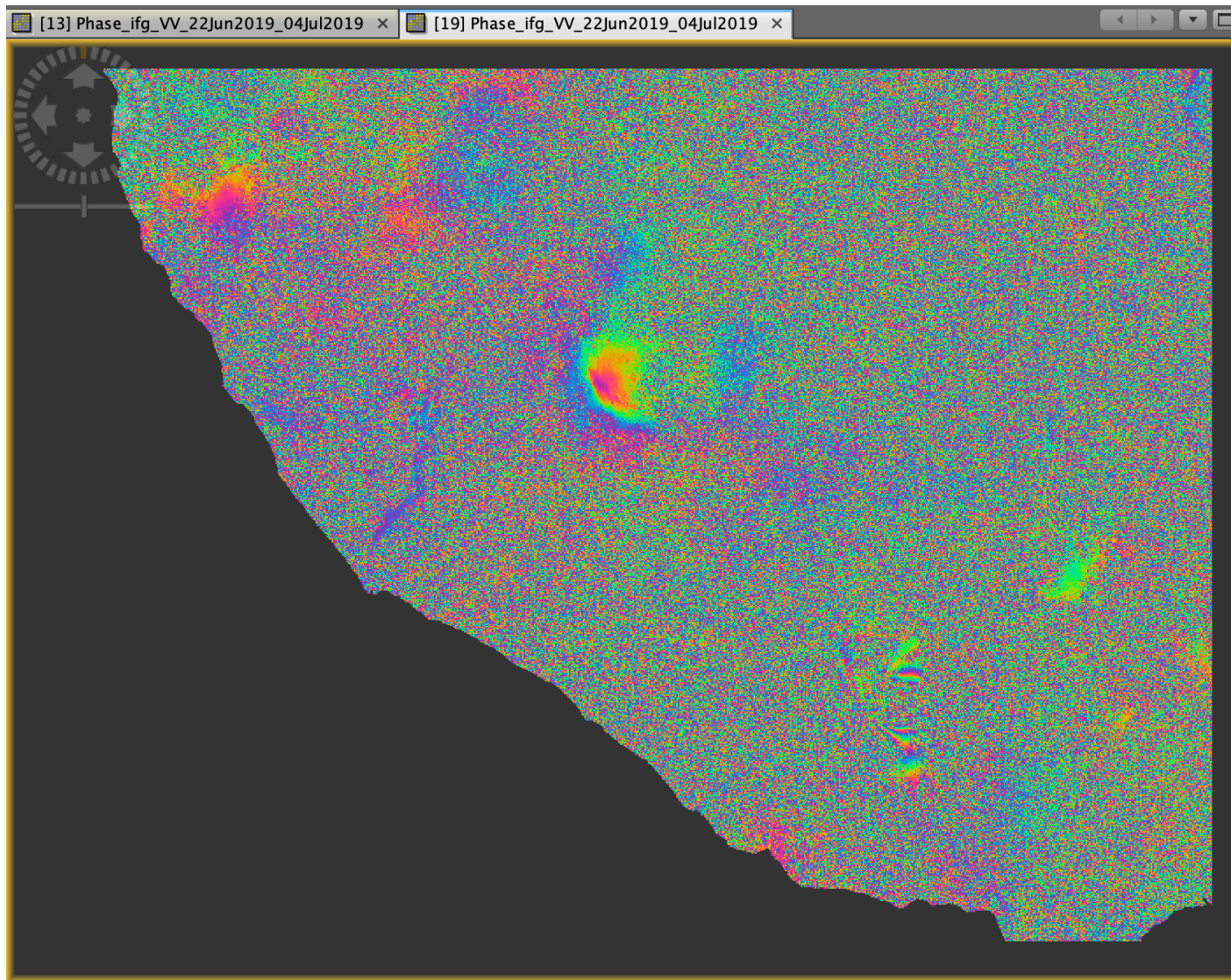
4.3. Goldstein Phase Filtering

The filtering (Radar > Interferometric > Filtering > Goldstein Phase Filtering) is carried out to reduce noise and improve the quality of phase data by removing the random phase error in interferogram. The coherence threshold ranges from 0 to 1, with values closer to 1 indicating high correlation, which corresponds to stable surfaces.





This is the result after applying filter with coherence threshold 0.9.



This is the result after applying filter with coherence threshold 0.2.
Images are noisy even after applying the 0.9. But for the rest of the analysis, output from 0.9 value is used.

6. Creation of Displacement Map

6.1. Manual SNAPHU Installation

The SNAPHU plugin was downloaded from the <https://step.esa.int/main/snap-supported-plugins/snaphu/> for the windows operating system.

  → Science Toolbox Exploitation Platform 

Manual SNAPHU installation

Important note: The installation package for Linux was updated on 12.03.2018 9:05 UTC.
The installation file for version 1.4.2 is provided for 32- and 64-bit Windows OS and Linux OS:

- Installation file for 64-bit Windows: [snaphu-snaphu-v1.4.2_win64.zip](#)
- Installation file for 32-bit Windows: [snaphu-v1.4.2_win32.zip](#)
- Installation file for Linux: [snaphu-v1.4.2_linux.zip](#)

The Windows binaries were built using the MSYS2 environment. For both Windows and Linux, make and gcc packages were necessary. The last step was running the make command on the sources downloaded from [here](#).

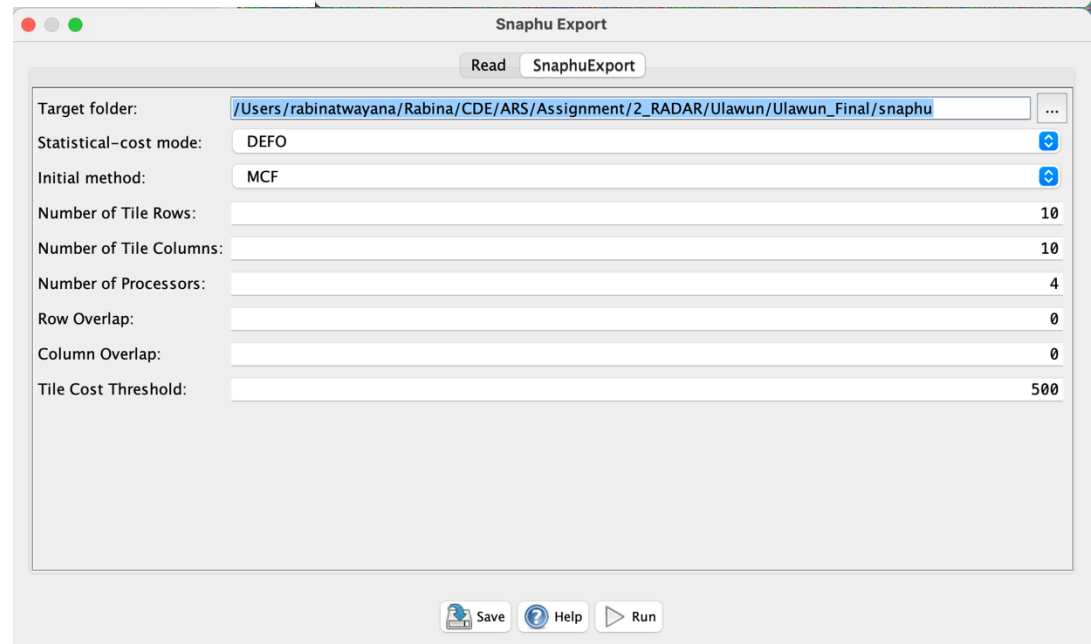
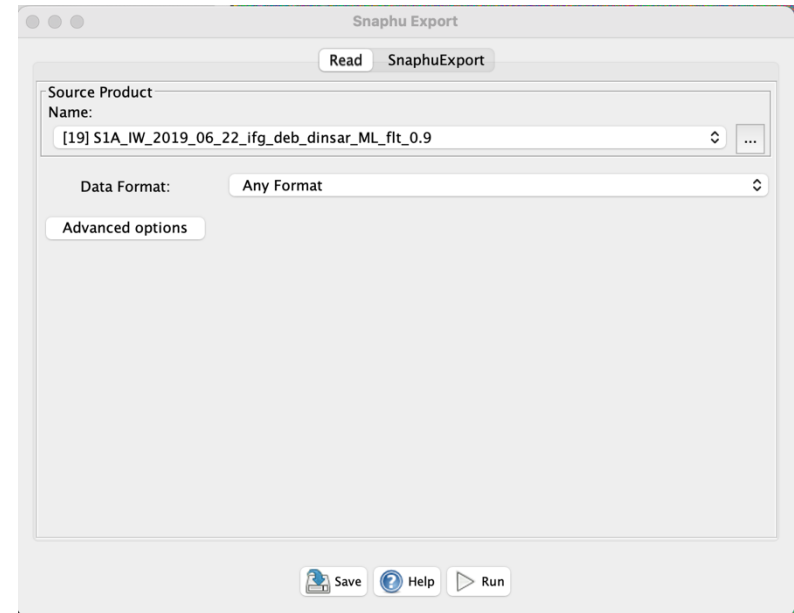
Installation and usage instructions:

1. Download the archive provided above and unzip it anywhere (preferably under your home folder)
2. Add the bin subfolder of the extracted folder in the system path.
3. Test your installation by running snaphu in a console. It should print the execution parameters.
4. Export the filtered flattened interferogram from SNAP to SNAPHU (Radar->Interferometric->Unwrapping->Snaphu Export)
5. Step into the output folder, where you should find the snaphu.conf generated file
6. Run SNAPHU from the console using the command which can be found in snaphu.conf (file is in the folder where the wrapped phase was exported) e.g:
`snaphu -f snaphu.conf Phase_VV_15Aug2016_27Aug2016.snaphu.img 6238`

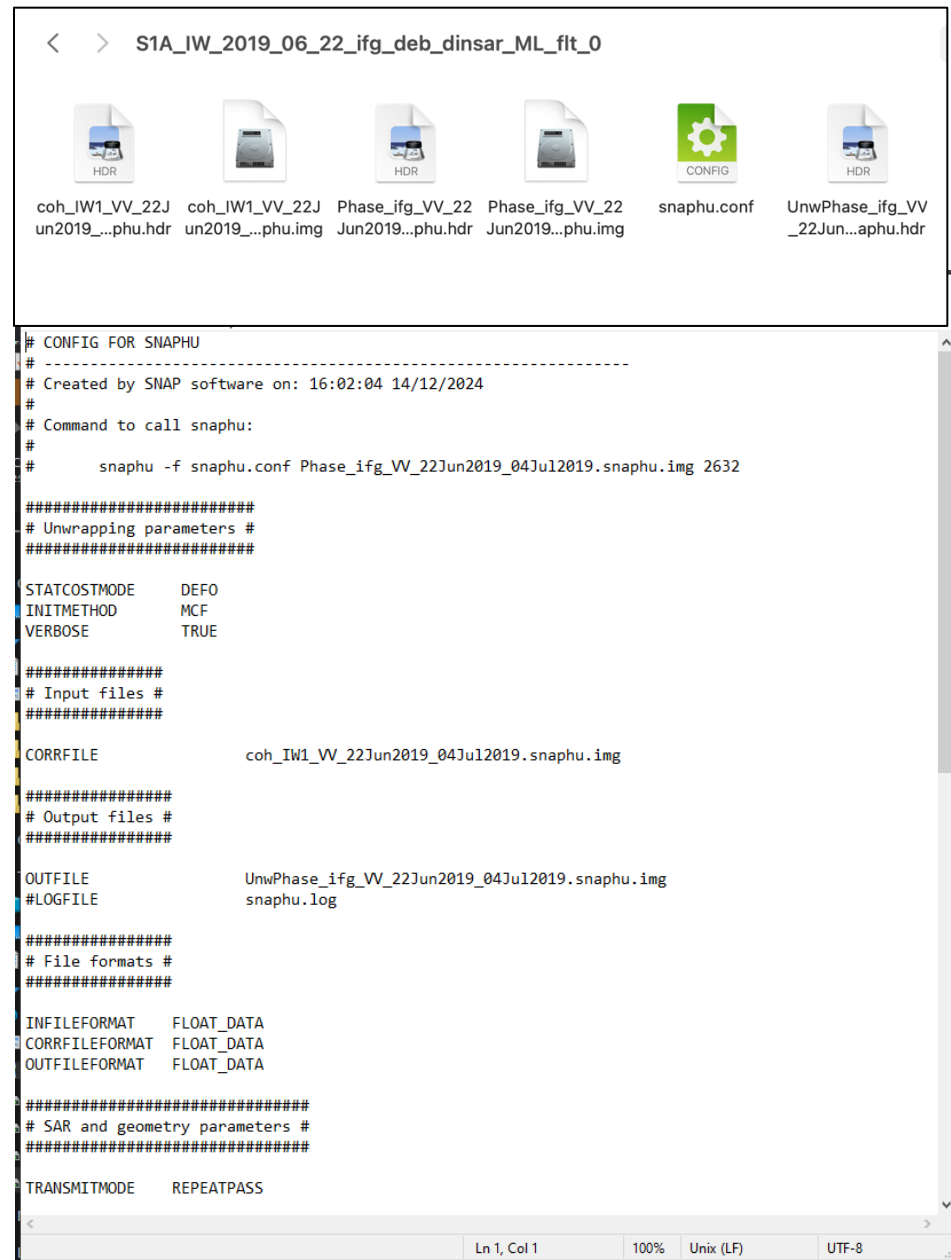
After the phase is unwrapped it can be imported to SNAP for further analysis using Radar->Interferometric->Unwrapping-> SNAPHU Import.

6.2..SNAPHU Export

In this step, we created the environment for snaphu. The following operation were being done: Radar > Interferometric > Unwrapping > SNAPHU Export. In the Snaphu Export window, we need to create a new folder for the snaphu-export output, and in the parameters we did not changed any of the values except the Number of Processors to 4. In this case, leaving the Number of Tile Rows and Columns on 10, tiles being processed in parallel and finishing this step quicker.



After navigating to the newly created snaphu folder, snaphu.conf file were changed by opening it with the notepad application. For the LOGFILE it needed to be commented out like #LOGFILE. Then the coherence file's name from the snaphu folder should be copied to match the name in CORRFILE line under the #Input file section. This modification needs to be saved before the next step.



The screenshot shows a file explorer window at the top with the title "S1A_IW_2019_06_22_ifg_deb_dinsar_MLflt_0". It contains several files: two HDR files (coh_IW1_VV_22Jun2019...phd.hdr, coh_IW1_VV_22Jun2019...phd.img), two Phase_ifg_VV_22Jun2019...phd.hdr files, a snaphu.conf file, and an UnwPhase_ifg_VV_22Jun2019...aphd.hdr file. Below the file explorer is a text editor window displaying the contents of the snaphu.conf file. The file is a configuration file for SNAP software, created on 14/12/2024. It includes sections for unwrapping parameters, input files, output files, file formats, and SAR and geometry parameters. The CORRFILE line is set to "coh_IW1_VV_22Jun2019_04Jul2019.snaphu.img" and the LOGFILE line is commented out as "#LOGFILE".

```
# CONFIG FOR SNAPHU
# -----
# Created by SNAP software on: 16:02:04 14/12/2024
#
# Command to call snaphu:
#
#       snaphu -f snaphu.conf Phase_ifg_VV_22Jun2019_04Jul2019.snaphu.img 2632
#
#####
# Unwrapping parameters #
#####

STATCOSTMODE      DEFO
INITMETHOD        MCF
VERBOSE           TRUE

#####
# Input files #
#####

CORRFILE           coh_IW1_VV_22Jun2019_04Jul2019.snaphu.img

#####
# Output files #
#####

OUTFILE            UnwPhase_ifg_VV_22Jun2019_04Jul2019.snaphu.img
#LOGFILE           snaphu.log

#####
# File formats #
#####

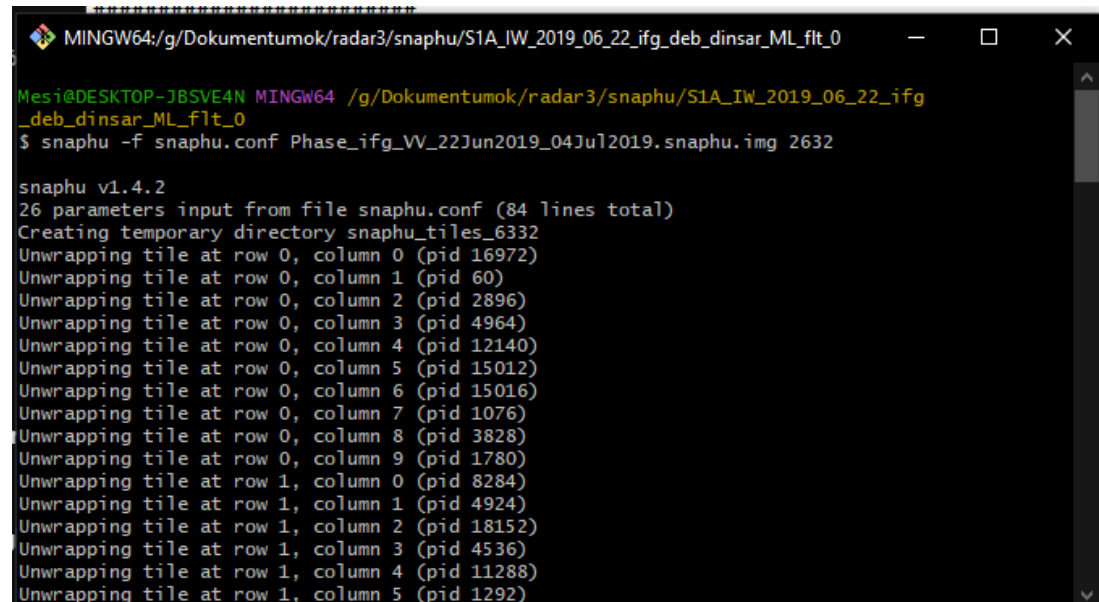
INFILEFORMAT       FLOAT_DATA
CORRFILEFORMAT     FLOAT_DATA
OUTFILEFORMAT      FLOAT_DATA

#####
# SAR and geometry parameters #
#####

TRANSMITMODE       REPEATPASS
```

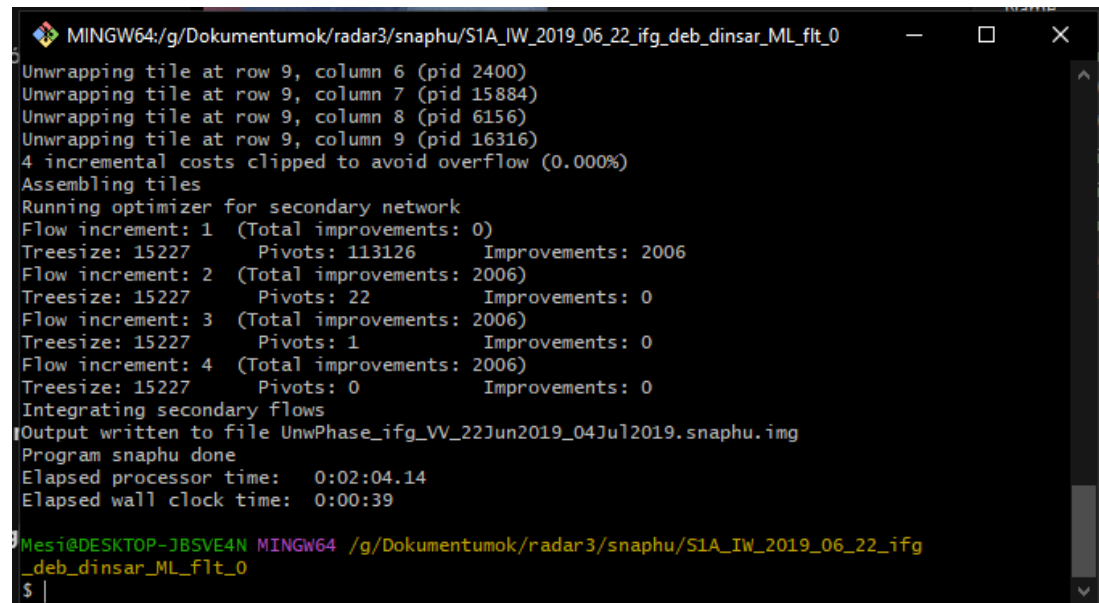
Then navigating back to the snaphu folder, we can run the command. We used the GitBash terminal for this process. which can be access by right clicking and clicked on open git bash here option

The process was completed with no error.



```
MINGW64: /g/Dokumentumok/radar3/snaphu/S1A_IW_2019_06_22_ifg_deb_dinsar_ML_ft_0
Mesidi@DESKTOP-JBSVE4N MINGW64 /g/Dokumentumok/radar3/snaphu/S1A_IW_2019_06_22_ifg_deb_dinsar_ML_ft_0
$ snaphu -f snaphu.conf Phase_ifg_VV_22Jun2019_04Jul2019.snaphu.img 2632

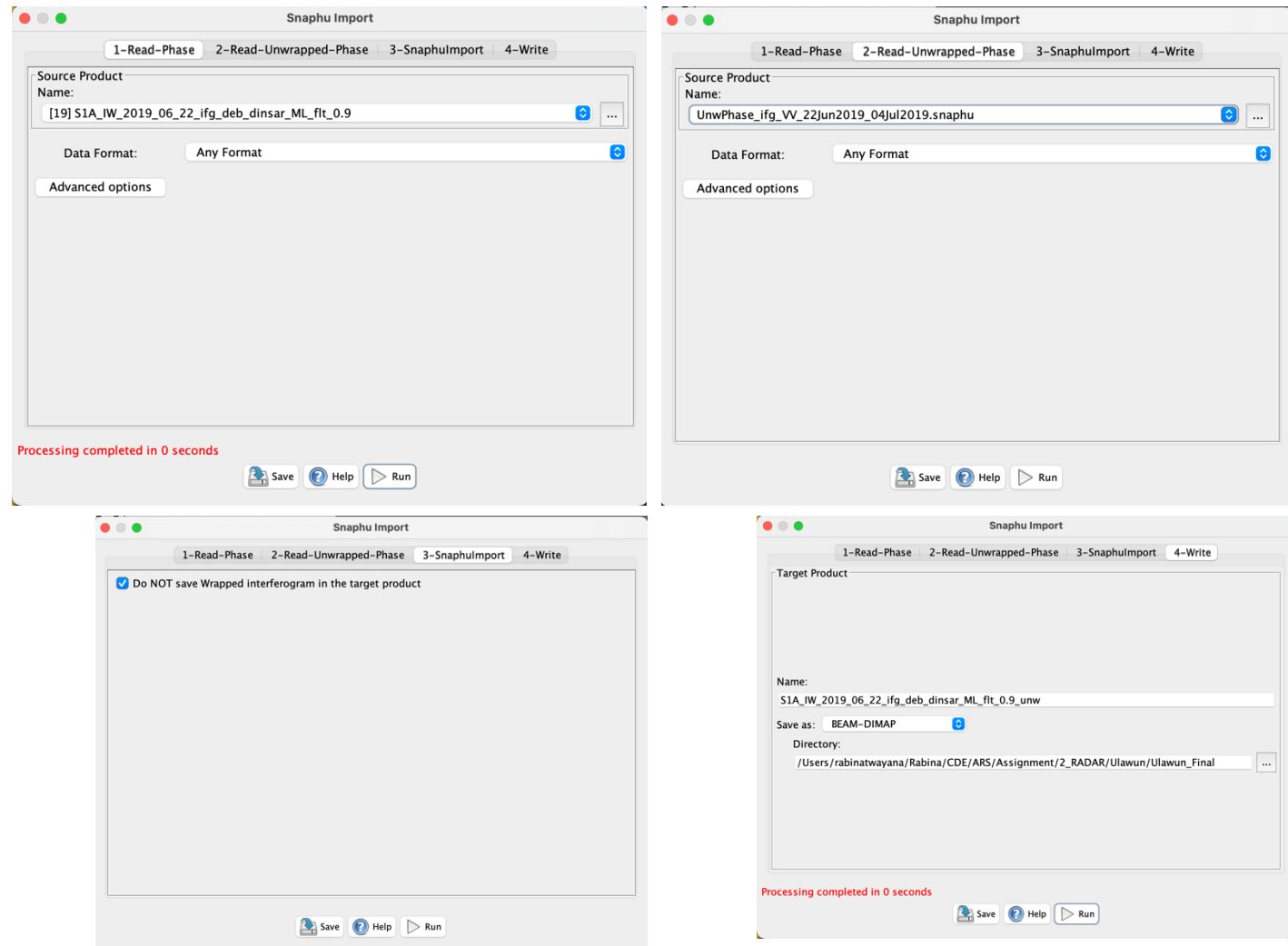
snaphu v1.4.2
26 parameters input from file snaphu.conf (84 lines total)
Creating temporary directory snaphu_tiles_6332
Unwrapping tile at row 0, column 0 (pid 16972)
Unwrapping tile at row 0, column 1 (pid 60)
Unwrapping tile at row 0, column 2 (pid 2896)
Unwrapping tile at row 0, column 3 (pid 4964)
Unwrapping tile at row 0, column 4 (pid 12140)
Unwrapping tile at row 0, column 5 (pid 15012)
Unwrapping tile at row 0, column 6 (pid 15016)
Unwrapping tile at row 0, column 7 (pid 1076)
Unwrapping tile at row 0, column 8 (pid 3828)
Unwrapping tile at row 0, column 9 (pid 1780)
Unwrapping tile at row 1, column 0 (pid 8284)
Unwrapping tile at row 1, column 1 (pid 4924)
Unwrapping tile at row 1, column 2 (pid 18152)
Unwrapping tile at row 1, column 3 (pid 4536)
Unwrapping tile at row 1, column 4 (pid 11288)
Unwrapping tile at row 1, column 5 (pid 1292)
```



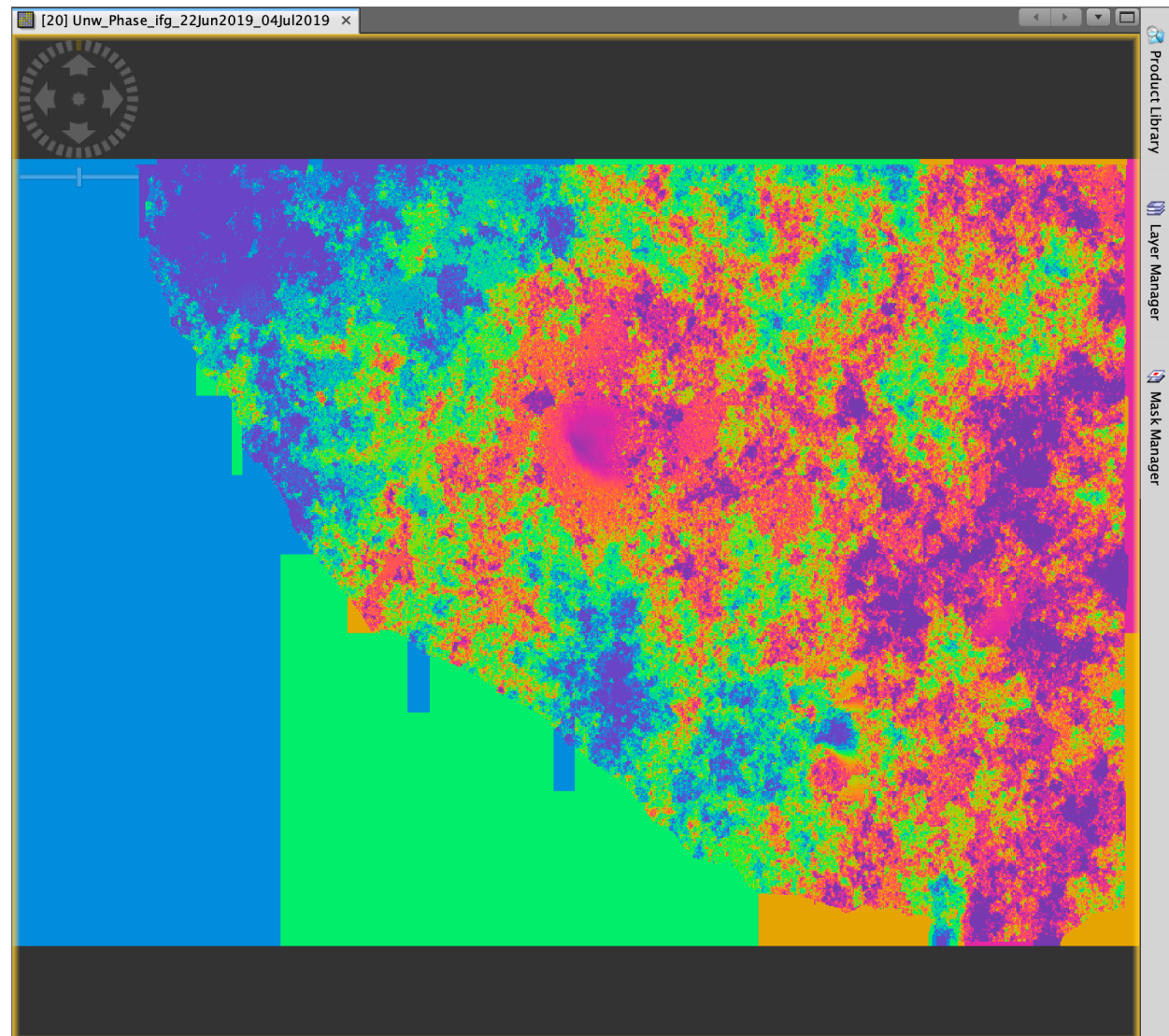
```
MINGW64: /g/Dokumentumok/radar3/snaphu/S1A_IW_2019_06_22_ifg_deb_dinsar_ML_ft_0
Unwrapping tile at row 9, column 6 (pid 2400)
Unwrapping tile at row 9, column 7 (pid 15884)
Unwrapping tile at row 9, column 8 (pid 6156)
Unwrapping tile at row 9, column 9 (pid 16316)
4 incremental costs clipped to avoid overflow (0.000%)
Assembling tiles
Running optimizer for secondary network
Flow increment: 1 (Total improvements: 0)
Treesize: 15227 Pivots: 113126 Improvements: 2006
Flow increment: 2 (Total improvements: 2006)
Treesize: 15227 Pivots: 22 Improvements: 0
Flow increment: 3 (Total improvements: 2006)
Treesize: 15227 Pivots: 1 Improvements: 0
Flow increment: 4 (Total improvements: 2006)
Treesize: 15227 Pivots: 0 Improvements: 0
Integrating secondary flows
Output written to file UnwPhase_ifg_VV_22Jun2019_04Jul2019.snaphu.img
Program snaphu done
Elapsed processor time: 0:02:04.14
Elapsed wall clock time: 0:00:39
Mesidi@DESKTOP-JBSVE4N MINGW64 /g/Dokumentumok/radar3/snaphu/S1A_IW_2019_06_22_ifg_deb_dinsar_ML_ft_0
$ |
```

6.3. SNAPHU Import

After the previous step, we need to navigate back to SNAP, then in the menu Radar > Interferometric > Unwrapping > Snaphu Import window we just need to select the correct output file and rename the new one with adding _unw at the end. Before running this process, the 'Don't save wrapped interferogram in the target Product' line was ticked. This will import the previously created new file into snaphu folder.

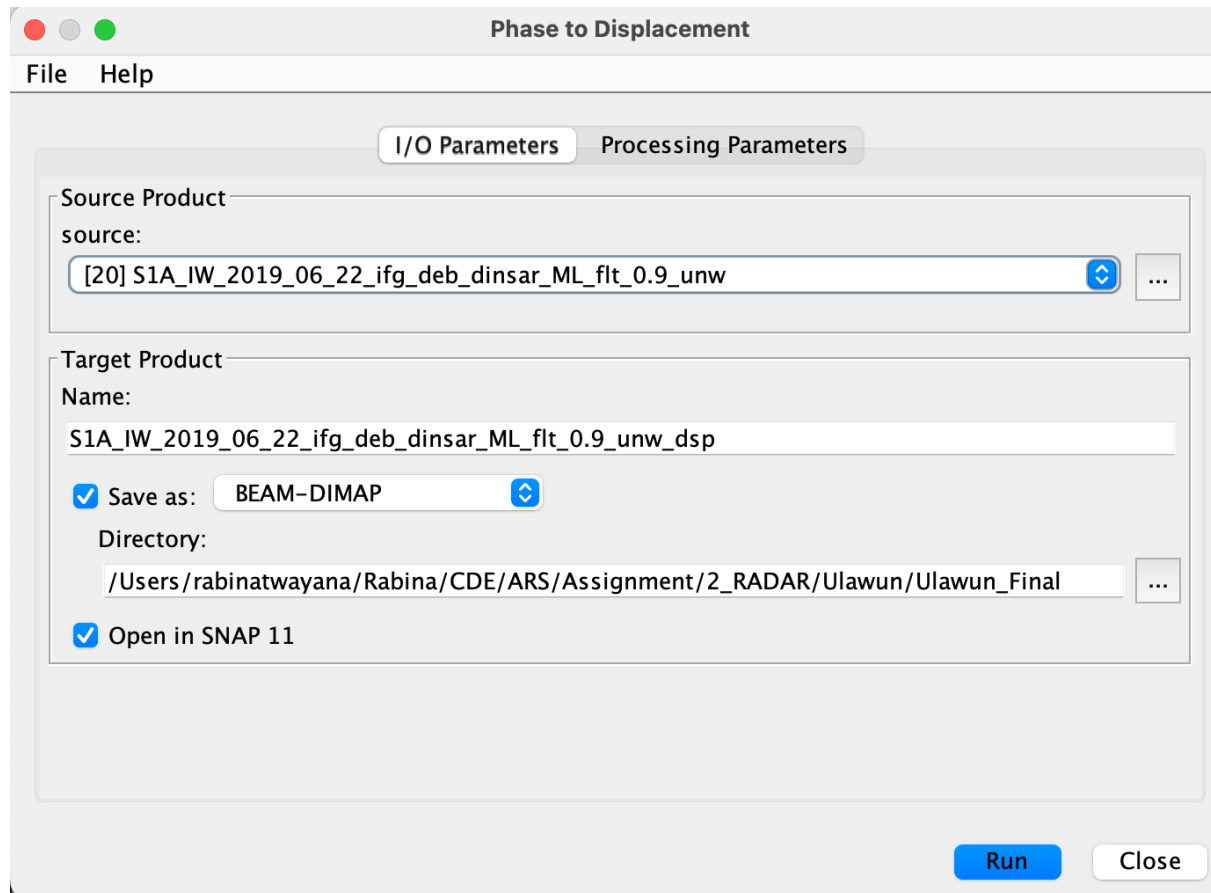


This will result in a new .hdr file, which can be visualised. In this step it's good to have a check on if the snaphu process was running correctly.

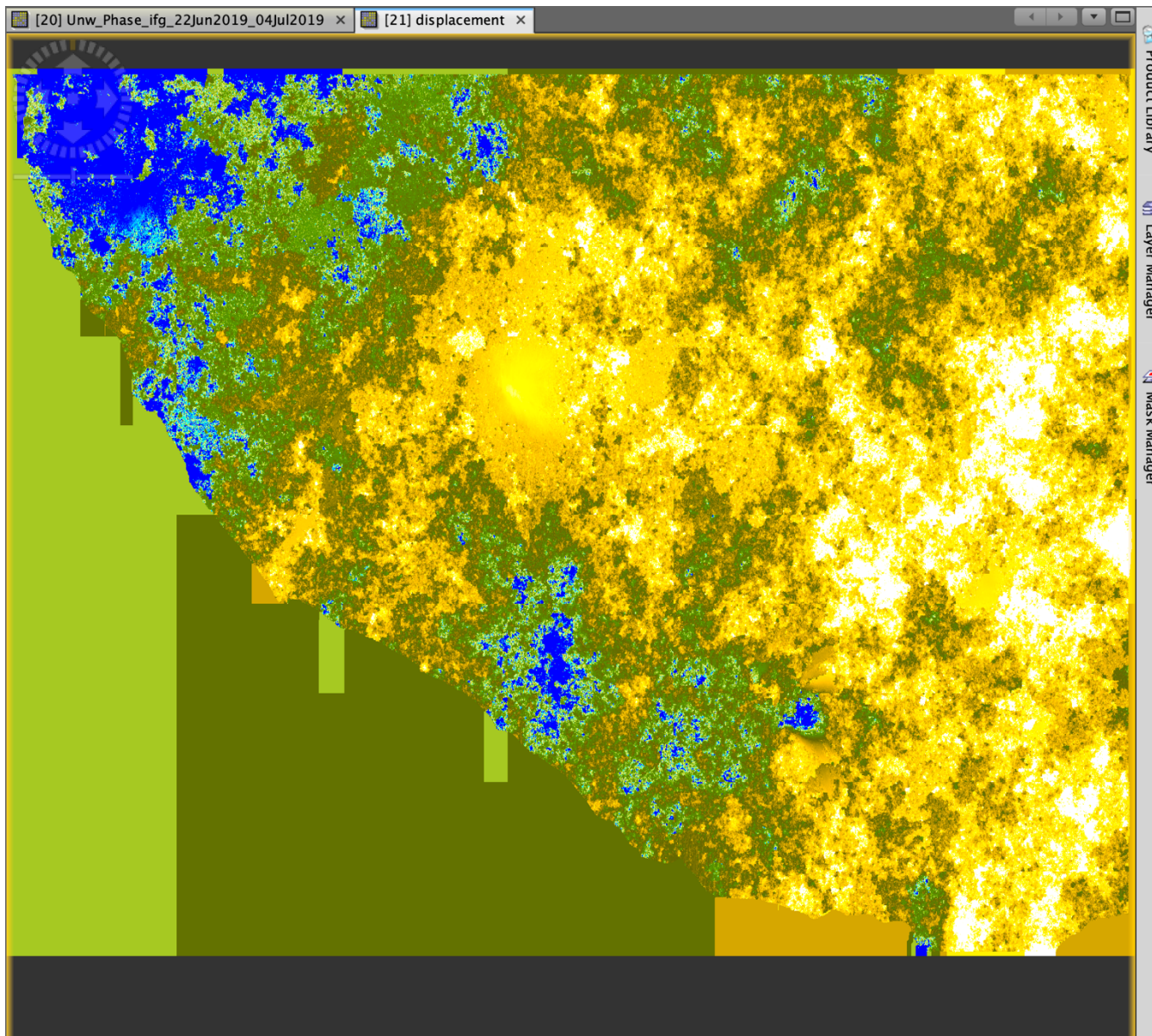


6.4. Phase to displacement

In this step the unwrapped phase images will be converted into displacement values, and this will let us see the results of ground movement, which can be analyzed. Radar > Interferometric > Products > Phase to Displacement window, there is nothing to change here so just run the process. We checked the results to make sure the process had run without mistake.

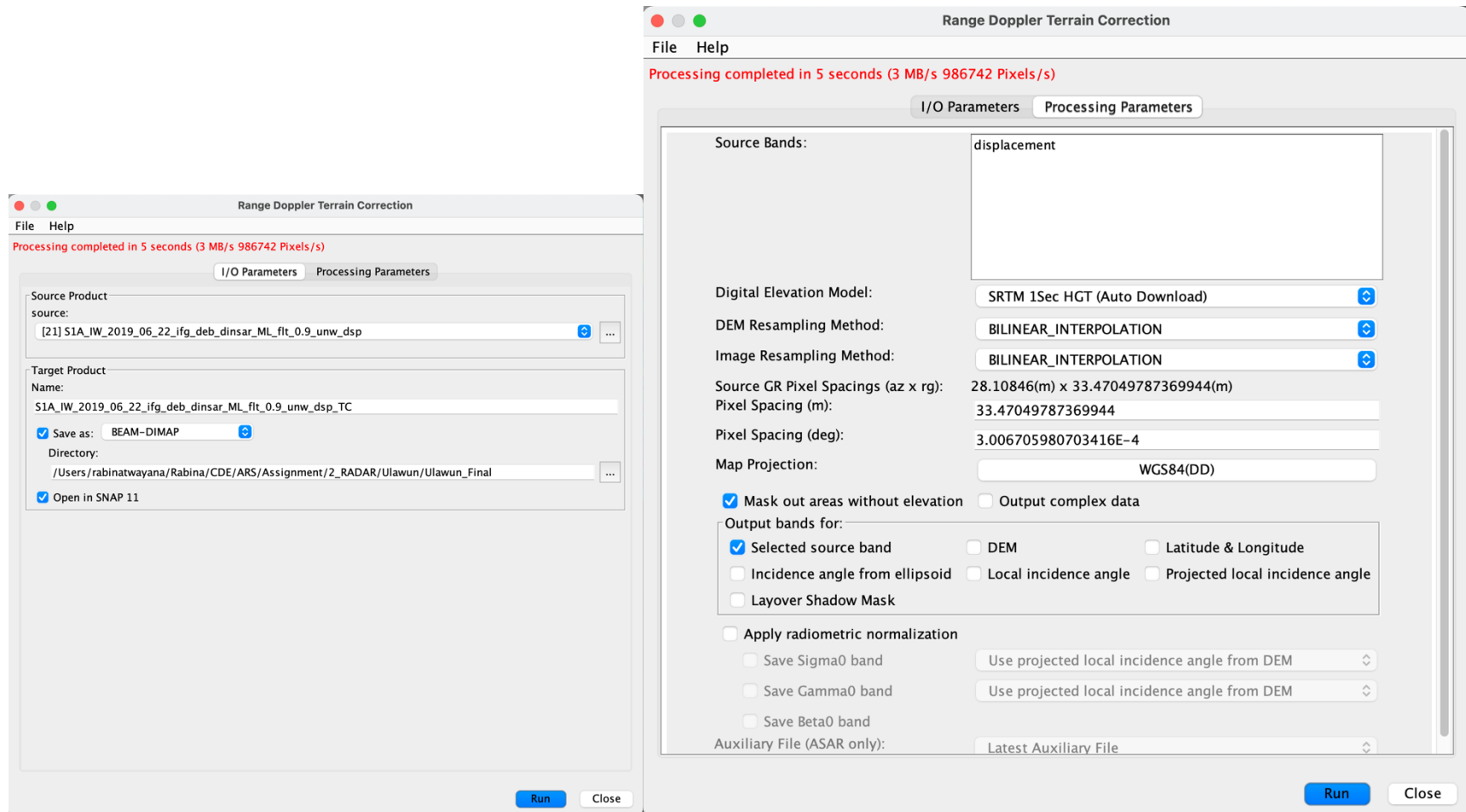


This is the displacement map.

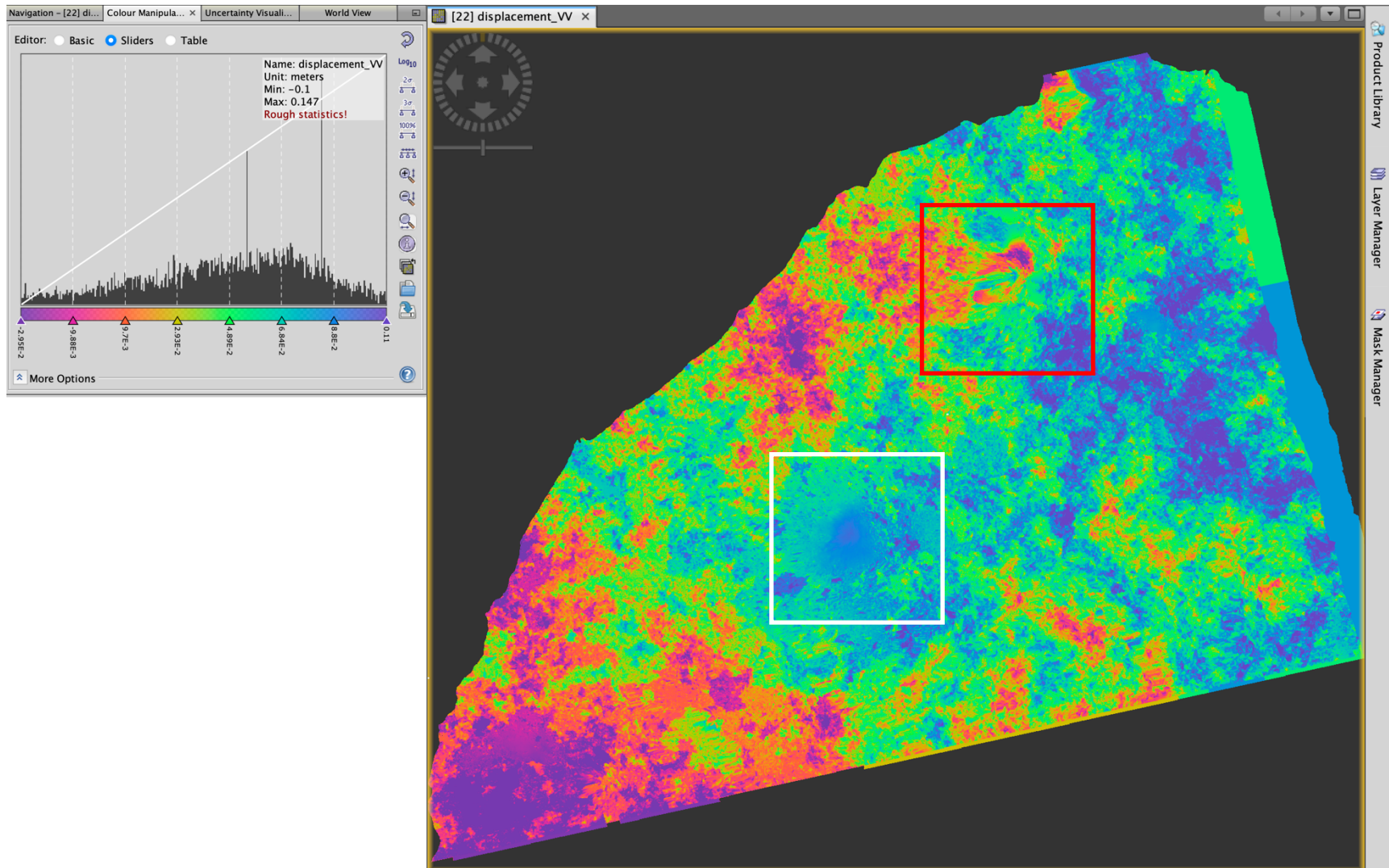


7. Terrain Correction

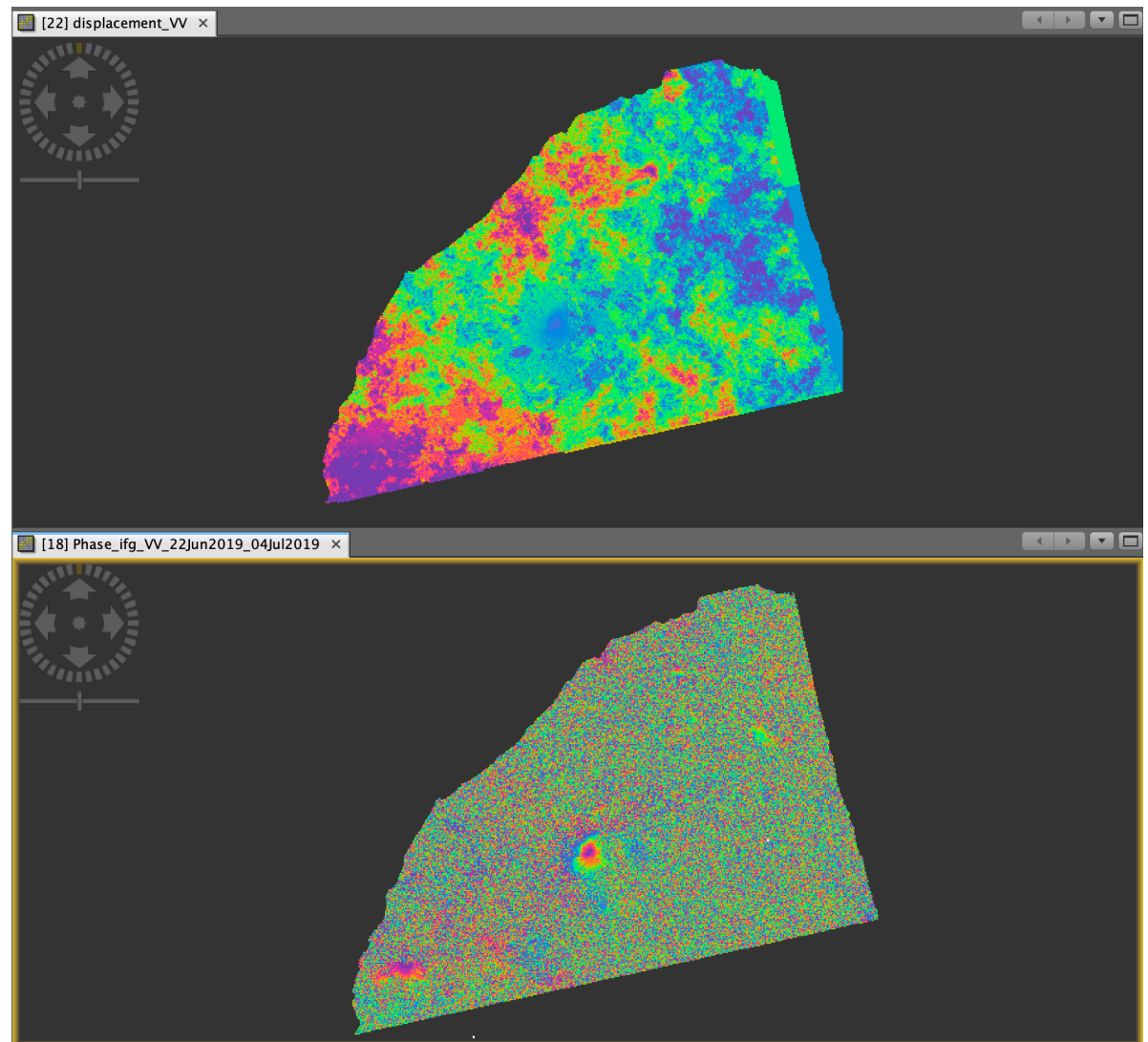
In this last step we need to correct our images by georeferencing them with the Terrain Correction tool. This process will put the displacement map into a coordinate system, at the same time removing any occurring distortions caused by topography. Radar > Geometric > Terrain Correction > Range-Doppler Terrain Correction. In the Processing Parameters tab, make sure to Select the correct DEM. We used SRTM 1Sec HGT.



Major event had happen in the region defined by red box in which small displacemnet can be witnessed for the selected study period with few number of fringes. In addition to this, displacement can also be seen in the region enclosed by white box with single fringe.

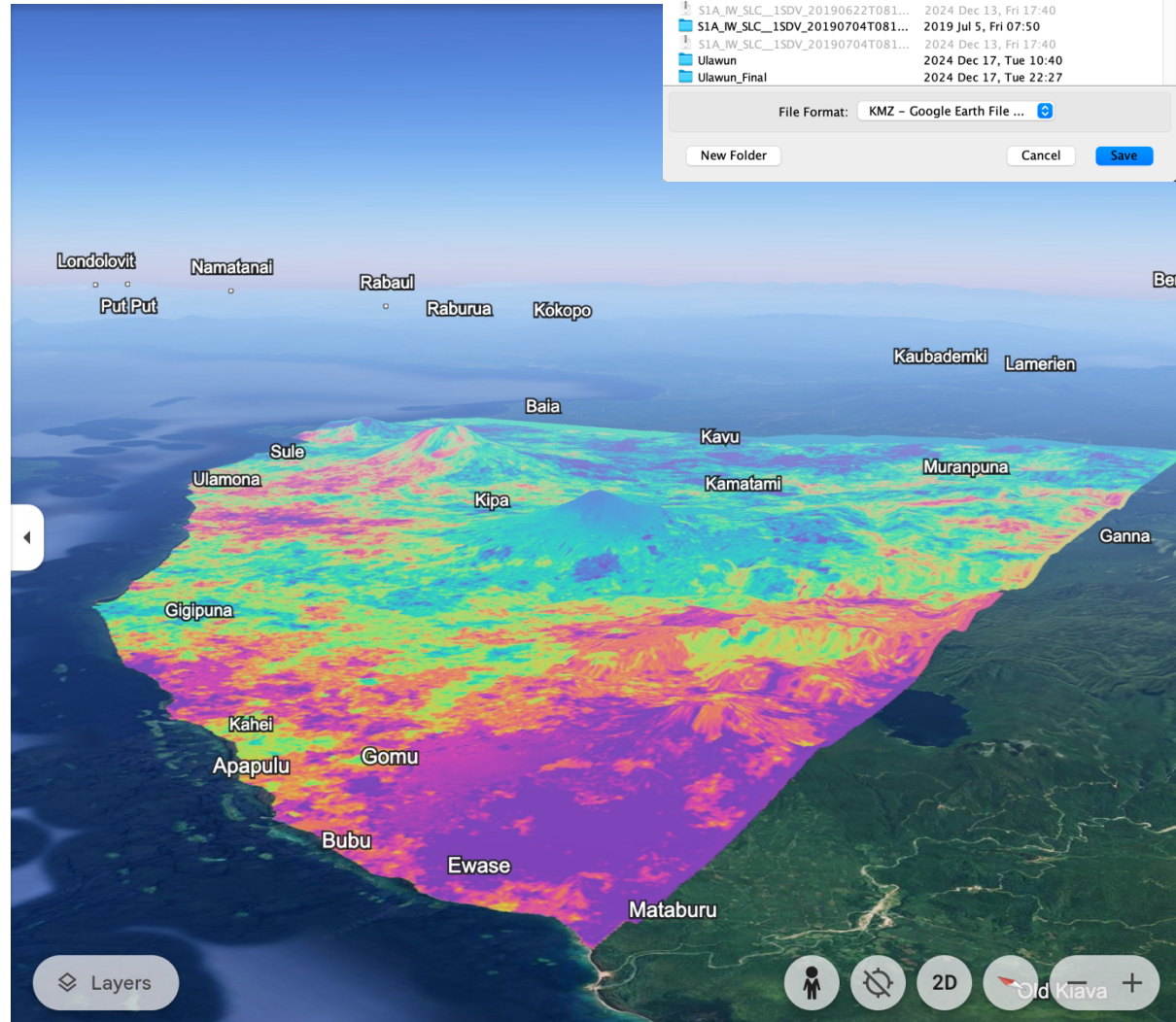


Terrain correction was applied similarly for the interferogram result obtained after Goldstein phase filtering. This is the comparison of topographic corrected displacement result and interferogram result.



8. Export Displacement Map

With a right click on the image, it can be exported in different format. We chose **Google Earth KMZ** for visualizing.



9. Conclusion

Volcano eruption can be mapped and analyzed effectively using SAR data with the help of the displacement map. This kind of analysis is important for detecting changes in the landscape due to the eruption. It can also be used for monitoring active volcanoes in time series. Additionally, it can also be used for damage assessment after the event by identifying areas that have been significantly altered by flows, or ash deposition. This can be useful for disaster management and recovery efforts. Goldstein phase filtering process was not much effective for filtering noise in our case study, thus alternative for this need to be explored.